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Shield Contact System

Abstract

A shield contact system includes a first shield and a second shield. The first shield has a first shield body with a first leading edge and a guiding element extending from the first leading edge along an insertion direction. The second shield has a second shield body defining a receiving space. The second shield body has a second leading edge. The first shield and the second shield are inserted together along the insertion direction to an inserted state in which the guiding element is positioned within the receiving space and the first leading edge is separated from the second leading edge along the insertion direction. The first leading edge does not overlap the second leading edge along the insertion direction in the inserted state.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates to a contact system and, more particularly, to a contact system connecting a first shield to a second shield.

BACKGROUND

[0002] In a shield contact system, a male shield is inserted into a female shield to form a connection between the shields. The female shield has a contact beam that extends into the space in which the male shield is received. The contact beam contacts an outer surface of the male shield to form the connection and slides along the outer surface of the male shield during insertion of the male shield.

[0003] During insertion, it can be difficult to align the male shield in the female shield, leading to stubbing of an end of the male shield on the end of the female shield and resulting in damage to the shields. Providing a larger radial gap between the male shield and the female shield can simplify the alignment in the initial mating, but allows for greater floating between the male shield and the female shield in the mated state, which decreases the stability of the electrical connection and can cause excessive wear on the contact surfaces. A male shield that is not properly aligned within the female shield and floats within the female shield leads to uneven deflection of the contact beams and poor connection between the shields. These factors limit the number of mating cycles in the usable life of the shield contact system.

SUMMARY

[0004] A shield contact system includes a first shield and a second shield. The first shield has a first shield body with a first leading edge and a guiding element extending from the first leading edge along an insertion direction. The second shield has a second shield body defining a receiving space. The second shield body has a second leading edge. The first shield and the second shield are inserted together along the insertion direction to an inserted state in which the guiding element is positioned within the receiving space and the first leading edge is separated from the second leading edge along the insertion direction. The first leading edge does not overlap the second leading edge along the insertion direction in the inserted state.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] The invention will now be described by way of example with reference to the accompanying Figures, of which:

[0006] FIG. 1 is a perspective view of a shield contact system according to an embodiment;

[0007] FIG. 2 is a perspective view of a first shield of the shield contact system;

[0008] FIG. 3 is a perspective view of a second shield of the shield contact system with a dielectric in the second shield;

[0009] FIG. 4 is a sectional side view of the shield contact system in an inserted state;

[0010] FIG. 5 is a sectional end view of the shield contact system in the inserted state;

[0011] FIG. 6 is a sectional side view of the shield contact system in a mated state; and

[0012] FIG. 7 is a sectional end view of the shield contact system in the mated state.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0013] Exemplary embodiments of the present disclosure will be described hereinafter in detail with reference to the attached drawings, wherein like reference numerals refer to like elements. The present disclosure may, however, be embodied in many different forms and should not be construed as being limited to the embodiments set forth herein; rather, these embodiments are provided so that the present disclosure will convey the concept of the disclosure to those skilled in the art. In addition, in the following detailed description, for purposes of explanation, numerous specific

details are set forth to provide a thorough understanding of the disclosed embodiments. However, it is apparent that one or more embodiments may also be implemented without these specific details. [0014] Throughout the drawings, only one of a plurality of identical elements may be labeled in a figure for clarity of the drawings, but the detailed description of the element herein applies equally to each of the identically appearing elements in the figure. Throughout the specification, directional descriptors are used such as “circumferential direction,” “insertion direction,” and “radial direction”. These descriptors are merely for clarity of the description and for differentiation of the various directions. These directional descriptors do not imply or require any particular orientation of the disclosed elements.

[0015] A shield contact system **10** according to an embodiment is shown in a mated state M in FIG. **1**. The shield contact system **10** includes a first shield **100** and a second shield **200** that is matable with the first shield **100** along an insertion direction I.

[0016] The first shield **100**, as shown in FIG. **2**, has a first shield body **110**, a plurality of guiding elements **120** extending from the first shield body **110**, and a plurality of ribs **130** extending from the first shield body **110**.

[0017] As shown in FIG. **2**, the first shield body **110** has a first exterior surface **112** and a first interior surface **114** opposite the first exterior surface **112** in a radial direction R perpendicular to the insertion direction I. The first shield body **110** has a first leading edge **116** at one end in the insertion direction I. In the shown embodiment, the first shield body **110** has a rounded- rectangular cross-sectional shape. In other embodiments, the first shield body **110** could have a circular cross-sectional shape, or any other cross-sectional shape used in shield contacts.

[0018] The guiding elements **120** each extend from the first leading edge **116** of the first shield body **110** in the insertion direction I, as shown in FIG. **2**. The guiding elements **120** each have a guiding end **122** opposite the first leading edge **116** in the insertion direction I. In the shown embodiment, the guiding elements **120** each have an approximately triangular shape and the guiding end **122** of the guiding elements **120** has a curved shape. In other embodiments, the guiding elements **120** could have other shapes and the guiding ends **122** may or may not be curved.

[0019] The guiding elements **120** each extend from the first leading edge **116** at a guiding angle **124** with respect to the first interior surface **114**, as shown in FIG. **4**. In the shown embodiment, the guiding angle **124** is an obtuse angle.

[0020] The guiding elements **120** are distributed in a circumferential direction C around the first shield body **110**. In the shown embodiment, four guiding elements **120** are distributed around the first shield body **110**. This embodiment is merely exemplary and, in other embodiments, less than four or more than four guiding elements **120** may be distributed around the first shield body **110** and may be arranged differently than in the positions shown in FIG. **2**.

[0021] The first shield body **110** has a chamfer **118** on the first leading edge **116** between the guiding elements **120** that are distributed around the first leading edge **116**, as shown in FIG. **2**. The chamfer **118** is angled between the first leading edge **116** and the first interior surface **114** of the first shield body **110**. In the shown embodiment, the chamfer **118** is formed on all portions of the first leading edge **116** between the guiding elements **120**. In other embodiments, the chamfer **118** may be formed on only some of the portions of the first leading edge **116** between the guiding elements **120**, or may be omitted.

[0022] The ribs **130** each extend from the first exterior surface **112** of the first shield body **110** in the radial direction R, as shown in FIG. **2**. The ribs **130** each have a leading end **132** in the insertion direction I. The leading ends **132** of the ribs **130** are each positioned adjacent to the first leading edge **116** but spaced apart from the first leading edge **116** along the insertion direction I. In the shown embodiment, the ribs **130** are each continuous along the insertion direction I. In other embodiments, the ribs **130** may each be a series of segments separated from one another along the insertion direction I.

[0023] The ribs **130** are distributed in the circumferential direction C around the first shield body

110. In the shown embodiment, six ribs **130** are distributed around the first shield body **110**. This embodiment is merely exemplary and, in other embodiments, less than six or more than six ribs **130** may be distributed around the first shield body **110** and may be arranged differently than in the positions shown in FIG. 2.

[0024] The first shield **100** is formed of a conductive material, such as aluminum. In the shown embodiment, the first shield **100** is monolithically formed in a single piece from the conductive material with the first shield body **110**, the guiding elements **120**, and the ribs **130**; in this embodiment, the guiding elements **120** may be formed by punching in a sheet with the first shield body **110**, the ribs **130** may be embossed on the first shield body **110**, and the first shield body **110** may be bent into the shape shown in FIG. 2. In other embodiments, the first shield body **110** may be formed separately and attached together with the guiding elements **120** and the ribs **130**.

[0025] The second shield **200**, as shown in FIG. 3, has a second shield body **210** and a plurality of contact beams **220** extending from the second shield body **210**.

[0026] As shown in FIG. 3, the second shield body **210** has a second exterior surface **212** and a second interior surface **214** opposite the second exterior surface **212** in the radial direction R. The second shield body **210** defines a receiving space **216** extending along the insertion direction I and has a plurality of beam openings **218** extending through the second shield body **210** in the radial direction R and communicating with the receiving space **216**. The second shield body **210** has a second leading edge **219** at one end in the insertion direction I. The second shield body **210** defines the receiving space **216** with a shape corresponding to the shape of the first shield body **110**. In the shown embodiment, the second shield body **210** defines the receiving space **216** with a rounded-rectangular cross-sectional shape in the shown embodiment. In other embodiments, the second shield body **210** could define the receiving space **216** as a circular cross-sectional shape, or any other cross-sectional shape corresponding to the first shield body **110** and used in shield contacts.

[0027] The contact beams **220** are each positioned in one of the beam openings **218**, as shown in FIG. 3. The contact beams **220**, in the shown embodiment, are each cantilevered from the second shield body **210** and extend from a fixed end **222** to a cantilevered end **224** along the insertion direction I. In the shown embodiment, the cantilevered end **224** is positioned further from the second leading edge **219** of the second shield body **210** than the fixed end **222** along the insertion direction I. In another embodiment, the contact beam **220** may extend in the reverse direction with the cantilevered end **224** positioned close to the second leading edge **219** than the fixed end **222** along the insertion direction I. In other embodiments, the contact beams **220** may be connected at both ends but otherwise function as described herein.

[0028] Each of the contact beams **220** has a beam body **230**, as shown in FIGS. 4 and 5. The beam body **230** has a first beam section **232** and a second beam section **234** connected to the first beam section **232** at a contact point **236**. The second beam section **234** is bent at an angle with respect to the first beam section **232**; the angle of the second beam section **234** with respect to the first beam section **232** is greater than 90° and less than 180°. The contact point **236** is positioned at the intersection of the first beam section **232** and the second beam section **234**. The contact point **236** is spaced apart from the cantilevered end **224** along the insertion direction I.

[0029] The contact beams **220**, as shown in FIG. 3, are distributed in the circumferential direction C around the second shield body **210**. In the shown embodiment, four contact beams **220** are distributed around the second shield body **210**. This embodiment is merely exemplary; the number of contact beams **220** of the second shield **200** and the location of the contact beams **220** on the second shield **200** in the circumferential direction C corresponds to the number of guiding elements **120** of the first shield **100** and the locations of the guiding elements **120** on the first shield **100** in the circumferential direction C.

[0030] The second shield **200** is formed of a conductive material, such as aluminum. In the shown embodiment, the second shield **200** is monolithically formed in a single piece from the conductive material with the second shield body **210** and the contact beams **220**. The second shield body **210**

may be formed in a sheet, with the contact beams **220** punched and embossed, and then bent into the shape shown in FIG. **3**. In other embodiments, the second shield body **210** may be formed separately and attached together with the contact beams **220**.

[0031] In the embodiment shown in FIG. **3**, a dielectric **240** is positioned within the receiving space **216** of the second shield **200**. The dielectric **240** has a mating face **242** facing opposite to the insertion direction I. The dielectric **240** has a plurality of cutouts **244** at sides of the mating face **242**. In the shown embodiment, the dielectric **240** has four cutouts **244** distributed around the dielectric **240** in the circumferential direction C. This embodiment is merely exemplary; the number of cutouts **244** and the location of the cutouts **244** on the dielectric **240** in the circumferential direction C corresponds to the number of guiding elements **120** of the first shield **100** and the locations of the guiding elements **120** on the first shield **100** in the circumferential direction C.

[0032] The mating of the first shield **100** and the second shield **200** to form the connection of the shield contact system **10** will now be described in greater detail primarily with respect to FIGS. **4-7**.

[0033] The first shield **100** and the second shield **200** are inserted together along the insertion direction I. The first shield **100** and the second shield **200** initially reach an inserted state S shown in FIG. **4** along the insertion direction I. In the inserted state S, the guiding features **120** are positioned within the receiving space **216** of the second shield body **210** and the first leading edge **116** is separated from the second leading edge **219** along the insertion direction I; the guiding features **120** enter the receiving space **216** before the leading edges **116**, **219** reach one another along the insertion direction I.

[0034] When the first shield **100** and the second shield **200** reach the inserted state S, the guiding features **120** align the first shield **100** within the second shield **200** for further insertion along the insertion direction I. The guiding angle **124** of the guiding elements **120** and the protrusion of the guiding elements **120** beyond the first leading edge **116** assist a user in positioning the guiding elements **120** within the receiving space **216** at the second leading edge **219** and prevent stubbing or direct contact between the leading edges **116**, **219** during insertion. In the inserted state S, as shown in FIG. **5**, the first shield body **110** is aligned by the guiding elements **120** within the receiving space **216** of the second shield body **210** and the first leading edge **116** does not overlap the second leading edge **219**.

[0035] As shown in FIGS. **4** and **5**, the guiding elements **120** have entered and are positioned within the cutouts **244** of the dielectric **240** in the inserted state S. The cutouts **244** permit the alignment provided by the guiding elements **120** without having the dielectric **240** impede the movement of the guiding elements **120** and the first shield **100** along the insertion direction I.

[0036] The first shield **100** and the second shield **200** are inserted further together along the insertion direction I from the inserted state S to a mated state M shown in FIGS. **1**, **6**, and **7**.

[0037] As the shields **100**, **200** move together between the inserted state S and the mated state M, the first leading edge **116** first enters the receiving space **216** at the second leading edge **219** and the ribs **130** then enter the receiving space **216**. The ribs **130** protruding from the first exterior surface **112** of the first shield body **110** are positioned between the first shield body **110** and the second shield body **210** as the shields **100**, **200** move together along the insertion direction I from a position after the inserted state S to the mated state M shown in FIG. **7**. The ribs **130** lessen the gap between the second interior surface **214** of the second shield body **210** and the first exterior surface **112** of the first shield body **110** in the radial direction R, providing stability between the connection between the first shield **100** and the second shield **200** in the mated state M and centering the first shield **100** in the receiving space **216**.

[0038] The guiding elements **120** are each aligned with one of the contact beams **220** along the insertion direction I, as shown in FIGS. **4-6**. From the inserted state S to the mated state M along the insertion direction I, the guiding elements **120** pass through the cutouts **244** of the dielectric **240**

and then initially contact the contact beams **220** at the fixed end **222**. The guiding elements **120** of the first shield **100** contact the contact beams **220** after aligning the first shield **100** within the second shield **200** as the shields **100**, **200** move together along the insertion direction I. [0039] As shown in FIGS. **4** and **6**, as the shields **100**, **200** move together along the insertion direction I, the guiding elements **120** each contact and move along one of the contact beams **220**, deflecting the contact beam **220** away from the first shield body **110** and away from the receiving space **216** in the radial direction R. The guiding end **122** of the guiding element **120** having a curved shape prevents damage to the contact beam **220** at initial contact and the guiding angle **124** of the guiding element **120** forms an angled outer surface for a gradual progressive deflection of the contact beam **220** as the first shield **100** moves along the insertion direction I. The shown embodiment of the contact beam **220** in which the cantilevered end **224** is positioned further from the second leading edge **219** than the fixed end **222** further limits damaging contact between the guiding element **120** and the contact beam **220** and provides for a more gradual deflection of the contact beam **220** by the first shield **100**.

[0040] The contact beam **220** is at a maximum deflection when the contact point **236** has passed the guiding element **120** and is in contact with the first exterior surface **112** of the first shield body **110**. As the shields **100**, **200** are further inserted together along the insertion direction I to the mated state M shown in FIG. **6**, the contact point **236** slides along the first exterior surface **112** of the first shield body **110** by a wiping distance **250**. In the mated state M, as shown in FIGS. **6** and **7**, the contact points **236** of the contact beams **220** abut the first exterior surface **112** of the first shield body **110** and form the connection between the first shield **100** and the second shield **200**.

[0041] The shield contact system **10** described above and shown in the drawings is one possible embodiment that incorporates the features of the invention. In another embodiment, for example, the ribs **130** can be positioned on the second shield **200** and extend from the second interior surface **214** into the receiving space **216**.

[0042] In the mated state M, the first shield **100** and the second shield **200** are electrically connected. The shield contact system **10** may be part of an electrical connector system that electrically connects electrical components. In this application, the first shield **100** is positioned around a first dielectric having a first contact and the dielectric **240** is a second dielectric in the second shield **200** that has a second contact. The first shield **100** and the second shield **200** move into the mated state M described above to provide electromagnetic shielding for the mating between the first contact and the second contact.

[0043] In the shield contact system **10** according to the embodiments described above, the guiding elements **120** provide alignment of the first shield **100** within the second shield **200** that cases mating of the shields **100**, **200** along the insertion direction I and prevents stubbing of the leading edges **116**, **219**. The ribs **130** lessen the radial gap between the shields **100**, **200** in the mated state M, which decreases floating between the shields **100**, **200** and provides centering, limiting wear from floating movement and providing for approximately even deflection of the contact beams **220**. The embodiment of the contacts beams **220** in which the cantilevered end **224** is positioned further from the second leading edge **219** than the fixed end **222** decreases the wiping distance **250** in comparison to an exemplary wiping distance D in FIG. **6** that would occur if the contact beam **220** was reversed, further limiting the wear on the shields **100**, **200**. The shield contact system **10** minimizes wear and allows for the shield contact system **10** to be used over a greater number of mating cycles.

Claims

1. A shield contact system, comprising: a first shield having a first shield body with a first leading edge and a guiding element extending from the first leading edge along an insertion direction; and a second shield having a second shield body defining a receiving space, the second shield body has a

second leading edge, the first shield and the second shield are inserted together along the insertion direction to an inserted state in which the guiding element is positioned within the receiving space and the first leading edge is separated from the second leading edge along the insertion direction, the first leading edge does not overlap the second leading edge along the insertion direction in the inserted state.

2. The shield contact system of claim 1, wherein the second shield has a contact beam extending from the second shield body.
3. The shield contact system of claim 2, wherein the first shield and the second shield are inserted together along the insertion direction from the inserted state to a mated state, the contact beam abuts the first shield body in the mated state.
4. The shield contact system of claim 3, wherein the guiding element contacts and deflects the contact beam during insertion along the insertion direction between the inserted state and the mated state.
5. The shield contact system of claim 2, wherein the guiding element is one of a plurality of guiding elements distributed around the first leading edge in a circumferential direction and the contact beam is one of a plurality of contact beams distributed around the second shield body in the circumferential direction.
6. The shield contact system of claim 5, wherein each of the guiding elements is aligned with one of the contact beams along the insertion direction.
7. The shield contact system of claim 1, wherein the guiding element extends from the first leading edge at an obtuse angle with respect to a first interior surface of the first shield body.
8. The shield contact system of claim 1, wherein a guiding end of the guiding element in the insertion direction has a curved shape.
9. The shield contact system of claim 1, wherein the guiding element has an approximately triangular shape.
10. The shield contact system of claim 1, wherein the guiding element is monolithically formed in a single piece with the first shield body.
11. The shield contact system of claim 1, wherein the guiding element is one of a plurality of guiding elements distributed around the first leading edge in a circumferential direction, the first shield body has a chamfer on the first leading edge between the guiding elements.
12. The shield contact system of claim 1, wherein the first shield has a rib extending from the first shield body, the rib is positioned between the first shield body and the second shield body in a mated state of the first shield and the second shield.
13. The shield contact system of claim 12, wherein a leading end of the rib is spaced apart from the first leading edge along the insertion direction.
14. The shield contact system of claim 2, wherein the contact beam is cantilevered from the second shield body and extends from a fixed end to a cantilevered end along the insertion direction.
15. The shield contact system of claim 14, wherein the cantilevered end is positioned further from a second leading edge of the second shield body than the fixed end along the insertion direction.
16. The shield contact system of claim 1, further comprising a dielectric positioned within the receiving space, the dielectric has a mating face with a cutout, the guiding element passes through the cutout during insertion along the insertion direction.
17. A process of connecting a shield contact system, comprising: providing a first shield having a first shield body with a first leading edge and a guiding element extending from the first leading edge along an insertion direction; providing a second shield having a second shield body defining a receiving space, the second shield body has a second leading edge; and inserting the first shield and the second shield together along the insertion direction to an inserted state in which the guiding element is positioned within the receiving space and the first leading edge is separated from the second leading edge along the insertion direction, the guiding element aligns the first shield within the second shield as the first shield and second shield are further inserted together along the

insertion direction from the inserted state.

18. The process of claim 17, wherein the first shield and the second shield are inserted together along the insertion direction from the inserted state to a mated state, the second shield has a contact beam extending from the second shield body and contacting the first shield body in the mated state, the guiding element aligns the first shield within the second shield during insertion along the insertion direction before the contact beam contacts the first shield body.

19. The process of claim 18, wherein the contact beam is cantilevered from the second shield body and extends from a fixed end to a cantilevered end along the insertion direction, the guiding element contacts the fixed end before the cantilevered end and deflects the contact beam during insertion from the inserted state to the mated state.

20. The process of claim 17, wherein the first shield has a rib extending from the first shield body, the rib centers the first shield within the receiving space.
